Contents


```
% ===============================================================
% DORMITORY WATER SUPPLY PUMP DESIGN
% Pumping water to elevated tank (70 m)
% Population = 500 to 600 people
% ===============================================================

clear; clc; close all;
```

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1. USER INPUTS

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```
people = 600;                  % number of residents
use_per_person = 300;          % liters/person/day

tank_height = 70;             % tank elevation [m]
tank_source = 10;             % water source level [m]

pump_hours = 8;               % pump runs per day [h]

pipe_length = 90;            % pipe length [m]
pipe_diameter = 0.20;         % pipe diameter [m]

K_minor = 8;                  % fittings / bends loss coefficient
epsilon = 4.5e-5;            % roughness [m]

% Water at 30°C
rho = 996;                    % density [kg/m^3]
g = 9.81;
mu = 0.8e-3;                  % dynamic viscosity [Pa.s]

eta_pump = 0.80;             % pump efficiency
eta_motor = 0.90;            % motor efficiency
Safety_margin = 1.15;        % safety factor
```

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2. DAILY DEMAND

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```
daily_L = people * use_per_person;      % liters/day
daily_m3 = daily_L / 1000;             % m^3/day

Q_m3h = daily_m3 / pump_hours;         % m^3/h
Q = Q_m3h / 3600;                     % m^3/s
```

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3. PIPE FLOW

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```
A = pi * pipe_diameter^2 / 4;
V = Q / A;

Re = rho * V * pipe_diameter / mu;

% Swamee-Jain friction factor

f = 0.25 / (log10(epsilon/(3.7*pipe_diameter) + 5.74/(Re^0.9)))^2;

hf = f * (pipe_length/pipe_diameter) * (V^2/(2*g));
hm = K_minor * (V^2/(2*g));

h_loss = hf + hm;
```

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4. TOTAL HEAD

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```
H_static = tank_height - tank_source;
H_total1 = H_static + h_loss;
H_total = Safety_margin * H_total1;
```

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5. PUMP SPEED ESTIMATION

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```
N = 2900;                      % suggested speed [rpm]
omega = 2*pi*N/60;

psi = 0.6;                   % head coefficient
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6. VELOCITY TRIANGLES

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```
% Impeller outlet sizing
U2 = sqrt(g * H_total / psi);
D2 = 2 * U2 / omega;

% Inlet diameter
D1 = 0.45 * D2;

% Inlet flow
Vr1 = 3.5;                     % selected radial velocity [m/s]
V_theta1 = 0;                 % no prewhirl
U1 = omega * D1 / 2;

Beta1 = rad2deg(atan(Vr1/U1));

b1 = Q / (pi * D1 * Vr1* psi);

% Outlet flow
Vr2 = Vr1;
b2 = Q / (pi * D2 * Vr2 * psi);

U2 = omega * D2 / 2;

Angle_Beta2 = 25;
Beta2 = deg2rad(Angle_Beta2);

V_theta2 = U2 - (Vr2 / tan(Beta2));
V2 = sqrt(Vr2^2 + V_theta2^2);

% Number of blade
z = 8.5 * sin(Beta2) / (1 - D1/D2);
```

```
% Pump theoretical head
H_pump = U2 * V_theta2 / g;

% Design ratio
y_p = H_total / H_pump;
```

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7. POWER

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```
% Euler theoretical shaft power
W_dot_shaft = rho * Q * (U2 * V_theta2 - U1 * V_theta1);

% Hydraulic power
P_hyd = rho * g * Q * H_total;

% Absorbed Poer
% Shaft power required
P_abs = H_total * Q_m3h * rho / (3.67e5 * eta_pump);

% Motor power required
P_motor = (P_abs / eta_motor) * 1.15;

DeltaP_t = rho * g * H_total;
DeltaP = DeltaP_t - 0.5 * rho * (V2^2 - Vr1^2);

% Specific speed
Ns = omega * sqrt(Q) /( (DeltaP_t / rho) ^0.75);

% Degree of reaction
R_rotor = 0.5 + (Q * cot(Beta2)) / (2 * pi * D2 * b2 * U2);
```

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8. RESULTS

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```
fprintf('\n--- Pressure Rise ---\n');
fprintf('Theoretical Shaft Power     : %.2f kW\n', W_dot_shaft/1000);
fprintf('Number of Blade             : %.0f Blades\n', round(z));
fprintf('Total Pressure Rise         : %.2f kPa\n', DeltaP_t/1000);
fprintf('Static Pressure Rise        : %.2f kPa\n', DeltaP/1000);
fprintf('Degree of Reaction          : %.2f\n', R_rotor);
fprintf('Specific Speed              : %.2f\n', Ns);
fprintf('y                           : %.2f \n', y_p);

fprintf('\n====================================================\n');
fprintf('      DORMITORY WATER SUPPLY PUMP DESIGN\n');
fprintf('====================================================\n');

fprintf('\n--- Population Data ---\n');
fprintf('People                     : %d\n', people);
fprintf('Use per person             : %.0f L/day\n', use_per_person);
fprintf('Daily demand               : %.2f m^3/day\n', daily_m3);

fprintf('\n--- Required Pump Flow ---\n');
fprintf('Pump operating hours       : %.1f h/day\n', pump_hours);
fprintf('Required flow              : %.2f m^3/h\n', Q_m3h);

fprintf('\n--- Pipe Analysis ---\n');
fprintf('Pipe diameter              : %.3f m\n', pipe_diameter);
fprintf('Velocity                   : %.2f m/s\n', V);
fprintf('Reynolds number            : %.2e\n', Re);
fprintf('Friction factor            : %.4f\n', f);

fprintf('\n--- Head Losses ---\n');
fprintf('Major loss hf              : %.2f m\n', hf);
fprintf('Minor loss hm              : %.2f m\n', hm);
fprintf('Total head loss            : %.2f m\n', h_loss);
fprintf('Static head                : %.2f m\n', H_static);
fprintf('Required total head        : %.2f m\n', H_total);
fprintf('Pump head                  : %.2f m\n', H_pump);

fprintf('\n--- Power ---\n');
fprintf('Hydraulic power            : %.2f kW\n', P_hyd/1000);
fprintf('Required shaft power       : %.2f kW\n', P_abs);
fprintf('Recommended motor          : %.2f kW\n', P_motor);


fprintf('\n--- Suggested Pump Design ---\n');
fprintf('Pump speed                 : %.0f rpm\n', N);

fprintf('\nInlet\n');
```

```
fprintf('D1                        : %.3f m\n', D1);
fprintf('Beta1                     : %.1f deg\n', Beta1);
fprintf('Width1                    : %.3f m\n', b1);

fprintf('\nOutlet\n');
fprintf('D2                        : %.3f m\n', D2);
fprintf('Beta2                     : %.1f deg\n', Angle_Beta2);
fprintf('Width2                    : %.3f m\n', b2);

fprintf('\n=====================================================\n');

% Check if the operating point meets your height requirement

if y_p < 1
    fprintf('Status        yb < 1          : SUCCESS   (System reaches 70m elevation)\n');
elseif y_p == 1
    fprintf('Status       yb = 1           : BE CAREFUL (Pump just meets the system head requirement)\n');
else
    fprintf('Status        yb > 1         : WARNING (Head insufficient for tank height)\n');
end
```

```
--- Pressure Rise ---
Theoretical Shaft Power     : 5.46 kW
Number of Blade             : 7 Blades
Total Pressure Rise         : 674.58 kPa
Static Pressure Rise        : 335.55 kPa
Degree of Reaction          : 0.57
Specific Speed              : 0.18
y                           : 0.77

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      DORMITORY WATER SUPPLY PUMP DESIGN
=====================================================

--- Population Data ---
People                     : 600
Use per person             : 300 L/day
Daily demand               : 180.00 m^3/day

--- Required Pump Flow ---
Pump operating hours       : 8.0 h/day
Required flow              : 22.50 m^3/h

--- Pipe Analysis ---
Pipe diameter              : 0.200 m
Velocity                   : 0.20 m/s
Reynolds number            : 4.95e+04
Friction factor            : 0.0217

--- Head Losses ---
Major loss hf              : 0.02 m
Minor loss hm              : 0.02 m
Total head loss            : 0.04 m
Static head                : 60.00 m
Required total head        : 69.04 m
Pump head                  : 89.36 m

--- Power ---
Hydraulic power            : 4.22 kW
Required shaft power       : 5.27 kW
Recommended motor          : 6.73 kW

--- Suggested Pump Design ---
Pump speed                 : 2900 rpm

Inlet
D1                         : 0.100 m
Beta1                      : 13.0 deg
Width1                     : 0.010 m

Outlet
D2                         : 0.221 m
Beta2                      : 25.0 deg
Width2                     : 0.004 m

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Status        yb < 1          : SUCCESS   (System reaches 70m elevation)
```

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9. INTEGRATED PUMP PERFORMANCE & RATIO ANALYSIS

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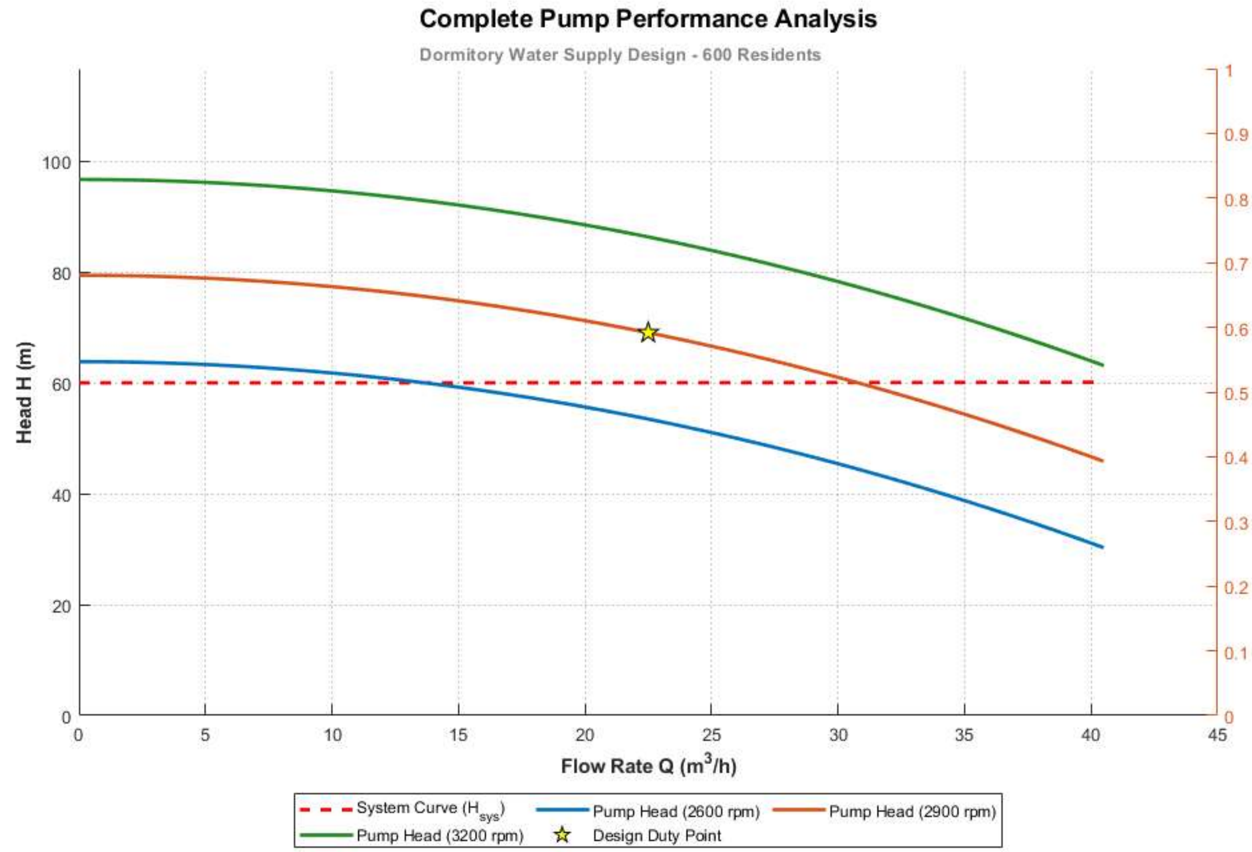
1. Setup X-axis (Flow Rate in m^3/h)

```matlab
Q_plot_m3h = linspace(0.001, 1.8 * Q_m3h, 250); % Avoid 0 for division
Q_plot_m3s = Q_plot_m3h / 3600;

% 2. System Curve (Hsys)
Ksys = h_loss / Q^2;
H_sys = H_static + Ksys .* Q_plot_m3s.^2;

% 3. Pump Curve Parameters (Affinity Laws)
N_design = 2900;
H0_design = Safety_margin * H_total;
Kp = (H0_design - H_total) / Q^2;

% Speeds and Styling
N_array = [2600, 2900, 3200];
colors = {[0 0.447 0.741], [0.850 0.325 0.098], [0.133 0.545 0.133]};

figure('Position', [100, 100, 1100, 700], 'Color', 'w');
hold on; grid on;
set(gca, 'GridLineStyle', ':', 'GridAlpha', 0.5);

% ==========================================
% LEFT Y-AXIS: Head (m)
% ==========================================
yyaxis left;
ylabel('Head H (m)', 'FontWeight', 'bold', 'FontSize', 11);
ylim([0, max(H0_design*(3200/2900)^2) + 20]);
ax = gca; ax.YColor = [0.15 0.15 0.15];

% Plot System Curve (Dashed Red)
plot(Q_plot_m3h, H_sys, '--r', 'LineWidth', 2, 'DisplayName', 'System Curve (H_{sys})');

% Plot Pump Curves (Solid Lines)
H_curves = cell(1, 3);
for i = 1:length(N_array)
    speed_ratio = N_array(i) / N_design;
    H0_speed = H0_design * (speed_ratio)^2;
    % Adjust Kp slightly for speed if using full affinity, but Kp is usually stable
    H_curve = H0_speed - Kp .* Q_plot_m3s.^2;
    H_curves{i} = H_curve;

    plot(Q_plot_m3h, H_curve, '-', 'Color', colors{i}, 'LineWidth', 2, ...
        'DisplayName', sprintf('Pump Head (%d rpm)', N_array(i)));
end


% ==========================================
% FINAL REFINEMENTS
% ==========================================
xlabel('Flow Rate Q (m^3/h)', 'FontWeight', 'bold', 'FontSize', 11);
title(['\fontsize{14}Complete Pump Performance Analysis\newline', ...
       '\fontsize{10}\color{gray}Dormitory Water Supply Design - 600 Residents'], 'FontWeight', 'bold');

% Legend placement
lgd = legend('Location', 'southoutside', 'Orientation', 'horizontal', 'NumColumns', 3);
lgd.Box = 'on';

% Add a point for the Design Duty Point
yyaxis left;
plot(Q_m3h, H_total, 'kp', 'MarkerSize', 12, 'MarkerFaceColor', 'y', 'DisplayName', 'Design Duty Point');
```



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